

2 Fig. 2.1 shows a gray tree frog, *Dryophytes versicolor*.

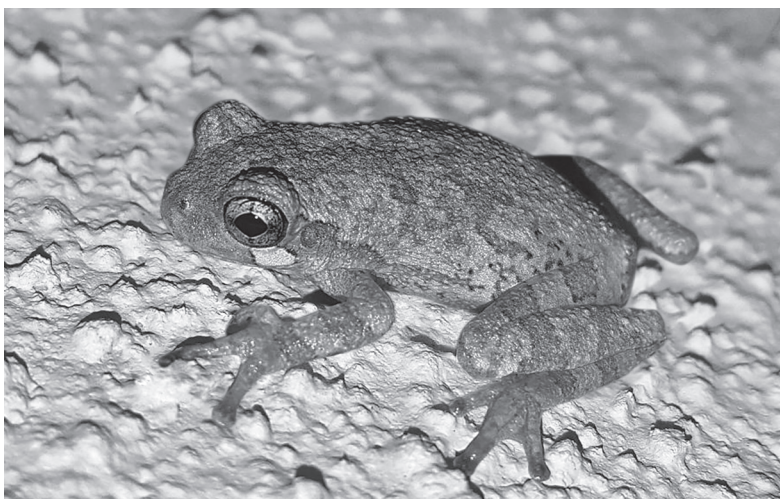


Fig. 2.1

Males of this species make a very distinctive mating call.

Some male gray tree frogs were kept in the laboratory at different temperatures. Their mating calls were recorded. Each call consists of a number of pulses. The calls are repeated at intervals.

Fig. 2.2 shows four calls recorded at 20 °C. Each vertical line represents a single pulse.

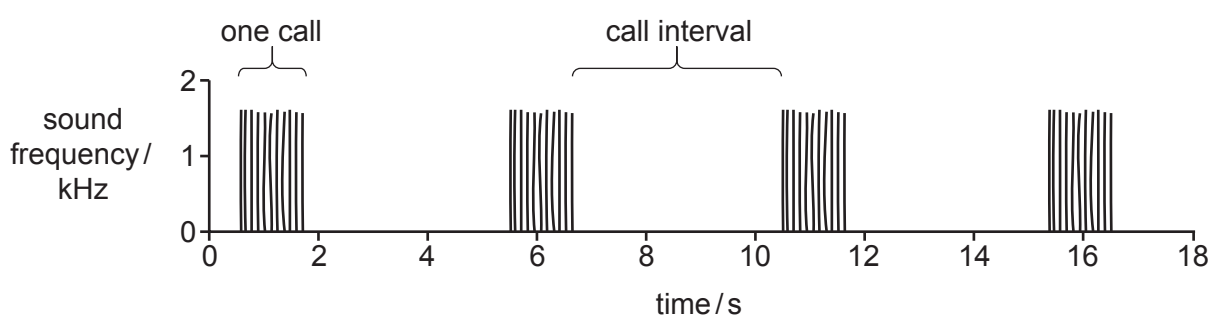


Fig. 2.2

A scientist wanted to find out if there was a relationship between body temperature and the interval between calls.

The scientist put each frog in a temperature-controlled environment. The scientist took the body temperature of each frog before recording the mating calls.

(a) State **one** variable, **other than** temperature, that should be standardised while making the recordings of the male gray tree frogs.

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..... [1]

(b) Recordings were made of 50 frogs with body temperatures between 9.0°C and 33.5°C.

The scientist measured the length of time of the intervals between the calls made by each frog. The results are shown in the scatter graph in Fig. 2.3.

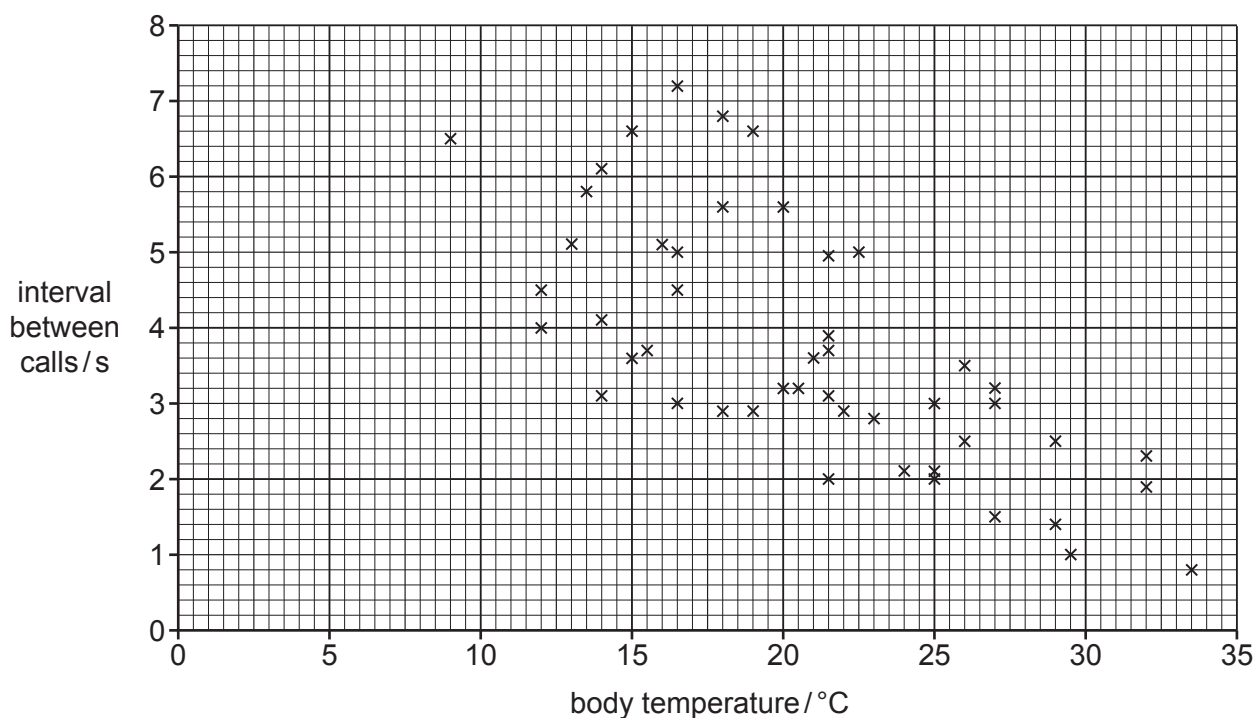


Fig. 2.3

Describe the trend shown by the scatter graph in Fig. 2.3.

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(c) The scientist analysed the data in Fig. 2.3 by using Pearson's linear correlation test.

(i) State **three** reasons why this statistical test is a suitable way to analyse the data.

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(ii) State a null hypothesis for this investigation.

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(iii) The scientist calculated the value of Pearson's correlation coefficient, r , to be **-0.717**.

Table 2.1 shows the probability table for Pearson's linear correlation test.

Table 2.1

numbers of pairs of measurements	critical values	
	$p = 0.05$ (5%)	$p = 0.01$ (1%)
48	0.285	0.368
49	0.282	0.365
50	0.279	0.361
51	0.276	0.358
52	0.273	0.354

Use the calculated value of r and Table 2.1 to explain whether the correlation is significant.

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- (d) The scientist extended the study by investigating other features of the mating calls of male gray tree frogs.

Fig. 2.4 shows recordings of the mating calls from two male gray tree frogs kept at different temperatures. Each vertical line represents a single pulse.

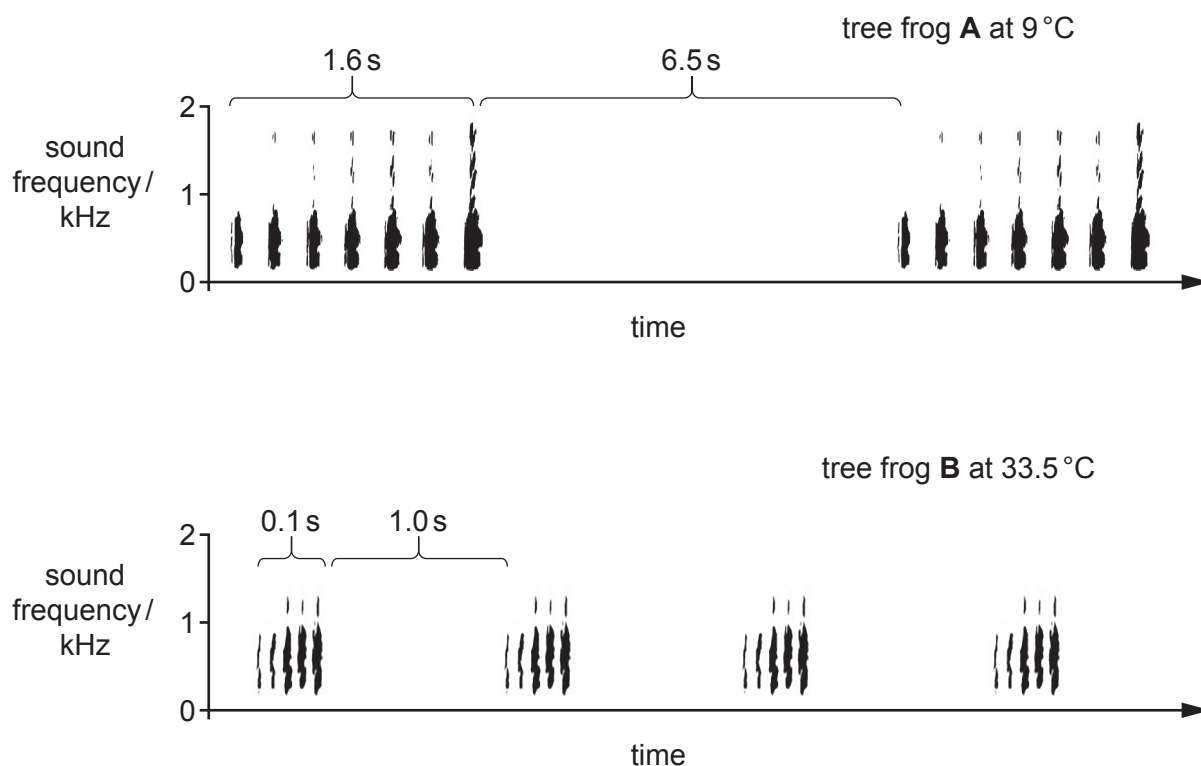


Fig. 2.4

Describe **two** features of the mating calls shown in Fig. 2.4, **other than** length of time between the calls, that could be studied.

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- (e) Other scientists have studied gray tree frogs in the wild. Their first task in these studies is to gain an estimate of the population in each study area.

Describe how a population of frogs, such as *D. versicolor*, could be estimated by using the mark-release-recapture method.

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..... [3]

[Total: 13]

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